

SQL Queries

Basic:

1. List all customers from USA.

```
select * from customers where country='USA';
```

output:

	AZ c	AZ company	AZ contact	AZ contact title	AZ address	AZ city	AZ state	AZ postal code	AZ country	AZ phone	AZ fax
2	HUNGC	Hungry Coyote Im	Yoshi Latimer	Sales Representative	City Center Plaza 516	Elgin	OR	97827	USA	(503) 555-6874	(503) 555-
3	LAZYK	Lazy K Kountry St	John Steel	Marketing Manager	12 Orchestra Terrace	Walla Wall	WA	99362	USA	(509) 555-7969	(509) 555-
4	LETSS	Let's Stop N Shop	Jaime Yorres	Owner	87 Polk St. Suite 5	San Francisco	CA	94117	USA	(415) 555-5938	[NULL]
5	LONEP	Lonesome Pine R	Fran Wilson	Sales Manager	89 Chiaroscuro Rd.	Portland	OR	97219	USA	(503) 555-9573	(503) 555-

2. List all products where UnitPrice is greater than 20.

```
select * from products where unitprice > 20;
```

output:

	AZ product	123 supplier	123 categoryid	AZ quantity per unit	123 unit price	123 units in stock	123 units on order
1	4	Chef Anton's C	2	48 - 6 oz jars	22	53	0
2	5	Chef Anton's G	2	36 boxes	21.35	0	0
3	6	Grandma's Boy	2	12 - 8 oz jars	25	120	0
4	7	Uncle Bob's Or	3	12 - 1 lb pkgs.	30	15	0

3. List all orders placed after 1997-01-01.

```
select * from orders where orderdate > '1997-01-01';
```

output:

select * from orders where orderdate > '1997-01-01' Enter a SQL expression to filter results (use Ctrl+Space)

	123 orderid	AZ customerid	123 employeeid	orderdate	requireddate	shippeddate	123 status
1	10,402	ERNSH	8	1997-01-02 00:00:00.000	1997-02-13 00:00:00.000	1997-01-10 00:00:00.000	2
2	10,403	ERNSH	4	1997-01-03 00:00:00.000	1997-01-31 00:00:00.000	1997-01-09 00:00:00.000	3
3	10,404	MAGAA	2	1997-01-03 00:00:00.000	1997-01-31 00:00:00.000	1997-01-08 00:00:00.000	1
4	10,405	LINOD	1	1997-01-06 00:00:00.000	1997-02-03 00:00:00.000	1997-01-22 00:00:00.000	1
5	10,406	QUEEN	7	1997-01-07 00:00:00.000	1997-02-18 00:00:00.000	1997-01-13 00:00:00.000	1

4. Display customers ordered by Country and then CompanyName.

select * from customers order by country, companyname;

output:

	AZ companyname	AZ contactnam	AZ contacttitl	AZ address	AZ city	AZ region	AZ postalcode	AZ country
1	CACTU Cactus Comidas para	Patricio Simpson	Sales Agent	Cerrito 333	Buenos Aires	[NULL]	1010	Argentina
2	OCEAN Océano Atlántico Ltda	Yvonne Moncada	Sales Agent	Ing. Gustavo Monca	Buenos Aires	[NULL]	1010	Argentina
3	RANCI Rancho grande	Sergio Gutiérrez	Sales Representa	Av. del Libertador 91	Buenos Aires	[NULL]	1010	Argentina
4	ERNSH Ernst Handel	Roland Mendel	Sales Manager	Kirchgasse 6	Graz	[NULL]	8010	Austria
5	PICCO Piccolo und mehr	Georg Pippas	Sales Manager	Geislweg 14	Salzburg	[NULL]	5020	Austria
6	MAISD Maison Dewey	Catherine Dewey	Sales Agent	Rue Joseph-Bens 5	Bruxelles	[NULL]	B-1180	Belgium
7	SUPRI Suprêmes délicies	Pascale Cartrain	Accounting Mana	Boulevard Tirou, 25	Charleroi	[NULL]	B-6000	Belgium
8	COMV Comércio Mineiro	Pedro Afonso	Sales Associate	Av. dos Lusíadas, 2	Sao Paulo	SP	05432-043	Brazil

5. List products ordered by highest UnitPrice first.

select * from products order by unitprice desc;

	productid	AZ productname	supplierid	categoryid	AZ quantityperunit	unitprice	unitsinstock
1	38	Côte de Blaye	18	1	12 - 75 cl bottles	263.5	
2	29	Thüringer Rostbratwurst	12	6	50 bags x 30 sausgs.	123.79	
3	9	Mishi Kobe Niku	4	6	18 - 500 g pkgs.	97	
4	20	Sir Rodney's Marmalade	8	3	30 gift boxes	81	
5	18	Carnarvon Tigers	7	8	16 kg pkg.	62.5	
6	59	Raclette Courdavault	28	4	5 kg pkg.	55	
7	51	Manjimup Dried Apples	24	7	50 - 300 g pkgs.	53	

Group By

6. Count how many customers are there in each country.

select country, count(*) as total_customers from customers group by country;

	AZ country	total_customers
1	Argentina	3
2	Spain	5
3	Switzerland	2
4	Italy	3
5	Venezuela	4
6	Belgium	2
7	Norway	1
8	Sweden	2
9	USA	13
10	France	11
11	Mexico	5

7. Find the number of products in each category.

select categoryid, **count**(*) **as** total_products **from** products **group by** categoryid;

	123 categoryid	123 total_products
1	8	12
2	7	5
3	1	12
4	5	7
5	4	10
6	2	12
7	6	6
8	3	12

8. Find the total number of orders handled by each employee.

select employeeid, **count**(*) **as** total_orders **from** orders **group by** employeeid;

	123 employeeid	123 total_orders
1	8	104
2	7	72
3	9	43
4	1	123
5	5	42
6	2	96
7	4	156
8	6	67
9	3	127

9. Find the average freight amount for each customer.

select customerid, **avg**(freight) **as** avg_freight_amount **from** orders **group by** customerid;

	A-Z customerid ▼	123 avg_freight_amount ▼
1	TOMSP	20.995
2	LONEP	9.9825
3	OLDWO	98.353
4	WARTH	54.832
5	MAGAA	46.975
6	QUEEN	152.5153846154
7	VINET	11.682

10. Find the maximum unit price **in each** category.

select categoryid, **max**(unitprice) **as** *max_unitprice* **from** products **group by** categoryid;

	123 categoryid ▼	123 max_unitprice ▼
1	8	62.5
2	7	53
3	1	263.5
4	5	38
5	4	55
6	2	43.9
7	6	123.79
8	3	81

HAVING

11. Show countries having more than 5 customers.

select country, **count**(*) **as** *total_customers* **from** customers **group by** country **having** **count**(*) > 5;

	A-Z country ▼	123 total_customers ▼
1	USA	13
2	France	11
3	Brazil	9
4	UK	7
5	Germany	11

12. Show employees who handled more than 50 orders.

```
select employeeid, count(*) as total_orders from orders group by employeeid
having count(*) > 50;
```

	123 employeeid	123 total_orders
1	8	104
2	7	72
3	1	123
4	2	96
5	4	156
6	6	67
7	3	127

13. Show customers whose average freight is greater than 50.

```
select customerid, round(avg(freight), 2) as avg_freight from orders group by
customerid having avg(freight) > 50;
```

	A-Z customerid	123 avg_freight
1	OLDWO	98.35
2	WARTH	54.83
3	QUEEN	152.52
4	MORGK	64.41
5	RATTC	118.57
6	FRANK	93.56
7	PRINI	72.97
8	LEHMS	67.8
9	RICAR	57.54
10	FOLIG	127.59
11	LINOD	56.15

14. Show categories where the average product price is greater than 30.

select categoryid, **avg**(unitprice) **as** avg_product_price **from** products **group by** categoryid **having** avg(unitprice) > 30;

	123 categoryid ▼	123 avg_product_price ▼
1	7	32.37
2	1	37.9791666667
3	6	54.0066666667

15. Show ship countries having more than 20 orders.

select shipcountry, **count**(*) **as** total_orders **from** orders **group by** shipcountry **having** count(*) > 20;

	A-Z shipcountry ▼	123 total_orders ▼
1	Spain	23
2	Italy	28
3	Venezuela	46
4	Sweden	37
5	USA	122
6	France	77

JOINS

16. List each order with customer company name.

```
select o.customerid, c.companyname, o.orderid from customers c join orders o
on o.customerid = c.customerid
```

	A-Z customerid ▼	A-Z companyname ▼	123 orderid ▼
1	VINET	Vins et alcools Chevalier	10,248
2	TOMSP	Toms Spezialitäten	10,249
3	HANAR	Hanari Carnes	10,250
4	VICTE	Victuailles en stock	10,251
5	SUPRD	Suprêmes délices	10,252
6	HANAR	Hanari Carnes	10,253
7	CHOPS	Chop-suey Chinese	10,254

17. List each order with employee first name and last name.

```
select e.employeeid, e.firstname, e.lastname, o.orderid from employees e join
orders o on e.employeeid = o.employeeid
```

	123 employeeid ▼	A-Z firstname ▼	A-Z lastname ▼	123 orderid ▼
1	5	Steven	Buchanan	10,248
2	6	Michael	Suyama	10,249
3	4	Margaret	Peacock	10,250
4	3	Janet	Leverling	10,251
5	4	Margaret	Peacock	10,252
6	3	Janet	Leverling	10,253

18. List products with their category name.

```
select p.productid, p.productname, c.categoryname from products p join
categories c on p.categoryid=c.categoryid;
```


	123 productid ▼	A-Z productname ▼	A-Z categoryname ▼
1	1	Chai	Beverages
2	2	Chang	Beverages
3	3	Aniseed Syrup	Condiments
4	4	Chef Anton's Cajun Seasoning	Condiments
5	5	Chef Anton's Gumbo Mix	Condiments
6	6	Grandma's Boysenberry Spread	Condiments
7	7	Uncle Bob's Organic Dried Pears	Produce
8	8	Northwoods Cranberry Sauce	Condiments

19. List products with supplier company name.

select *p*.productid, *p*.productname, *s*.companyname **as** *supplier_company* **from** products *p* **join** suppliers *s* **on** *p*.supplierid = *s*.supplierid;

	123 productid ▼	A-Z productname ▼	A-Z supplier_company ▼
1	1	Chai	Exotic Liquids
2	2	Chang	Exotic Liquids
3	3	Aniseed Syrup	Exotic Liquids
4	4	Chef Anton's Cajun Seasoning	New Orleans Cajun Delights
5	5	Chef Anton's Gumbo Mix	New Orleans Cajun Delights
6	6	Grandma's Boysenberry Spread	Grandma Kelly's Homestead

20. List orders with shipper company name.

select *o*.orderid, *s*.companyname **as** *shipper_company_name* **from** orders *o* **join** shippers *s* **on** *o*.shipvia = *s*.shipperid;

	123 orderid	A-Z shipper_company_name
1	10,248	Federal Shipping
2	10,249	Speedy Express
3	10,250	United Package
4	10,251	Speedy Express
5	10,252	United Package
6	10,253	United Package
7	10,254	United Package
8	10,255	Federal Shipping
9	10,256	United Package
10	10,257	Federal Shipping

MEDIUM

21. Find total orders per customer and display customer company name.

```
select o.customerid, c.companyname, count(*) as total_orders from customers c
join orders o on c.customerid = o.customerid group by o.customerid,
c.companyname;
```

	A-Z customerid	A-Z companyname	123 total_orders
1	SUPRD	Suprêmes délices	12
2	QUICK	QUICK-Stop	28
3	ANATR	Ana Trujillo Emparedados y helados	4
4	SAVEA	Save-a-lot Markets	31
5	TOMSP	Toms Spezialitäten	6
6	GALED	Galería del gastrónomo	5
7	LAUGB	Laughing Bacchus Wine Cellars	3
8	TRAIH	Trail's Head Gourmet Provisioners	3

22. Find total products supplied by each supplier.

```
select p.supplierid, s.contactname, count(*) as total_products_supplied from
products p join
suppliers s on s.supplierid = p.supplierid
group by p.supplierid, s.contactname;
```

	123 supplierid	A-Z contactname	123 total_products_supplied
1	24	Wendy Mackenzie	3
2	5	Antonio del Valle Saavedra	2
3	25	Jean-Guy Lauzon	2
4	19	Robb Merchant	2
5	14	Elio Rossi	3
6	1	Charlotte Cooper	3

23. Find average product price per category with category name.

```
select p.categoryid, c.categoryname, round(avg(unitprice), 2) as
avg_product_price from products p
join categories c on p.categoryid = c.categoryid group by p.categoryid,
c.categoryname;
```

	123 categoryid	A-Z categoryname	123 avg_product_price
1	8	Seafood	20.68
2	7	Produce	32.37
3	2	Condiments	23.06
4	3	Confections	25.16
5	4	Dairy Products	28.73
6	6	Meat/Poultry	54.01
7	5	Grains/Cereals	20.25
8	1	Beverages	37.98

24. Find total freight per customer and order by highest total freight.

```
select o.customerid, c.contactname, sum(freight) as total_freight from orders o
join customers c on c.customerid = o.customerid group by o.customerid,
c.contactname
order by total_freight desc;
```

	A-Z customerid	A-Z contactname	123 total_freight
1	SAVEA	Jose Pavarotti	6,683.7
2	ERNSH	Roland Mendel	6,205.39
3	QUICK	Horst Kloss	5,605.63
4	HUNGO	Patricia McKenna	2,755.24
5	RATTC	Paula Wilson	2,134.21
6	QUEEN	Lúcia Carvalho	1,982.7
7	FOLKO	Maria Larsson	1,678.08

25. Find employees who handled more than 25 orders.

```
select o.employeeid, e.firstname, e.lastname, count(*) as total_orders from orders
o
join employees e on e.employeeid = o.employeeid
group by o.employeeid, e.firstname, e.lastname
having count(*) > 25;
```

	123 employeeid	A-Z firstname	A-Z lastname	123 total_orders
1	8	Laura	Callahan	104
2	3	Janet	Leverling	127
3	6	Michael	Suyama	67
4	9	Anne	Dodsworth	43
5	7	Robert	King	72
6	1	Nancy	Davolio	123
7	4	Margaret	Peacock	156
8	5	Steven	Buchanan	42
9	2	Andrew	Fuller	96

ADVANCED:

26. Find total sales amount per order.

```
select orderid, sum(unitprice * quantity * (1-discount)) as total_sales_amount
from order_details od
group by orderid;
```

	123 orderid	123 total_sales_amount
1	11,038	732.5999997258
2	10,782	12.5
3	10,725	287.8
4	10,423	1,020
5	10,518	4,150.05
6	10,356	1,106.4
7	10,963	57.7999995947

27. Find total sales amount per customer.

```
select o.customerid, sum(unitprice * quantity * (1-discount)) as
total_sales_amount from order_details od
join orders o on o.orderid = od.orderid
join customers c on c.customerid = o.customerid
group by o.customerid;
```

	A-Z customerid	123 total_sales_amount
1	TOMSP	4,778.139997901
2	LONEP	4,258.6
3	OLDWO	15,177.4624818057
4	WARTH	15,648.7024694346
5	MAGAA	7,176.2149868105
6	QUEEN	25,717.4974730423

28. Find top 10 products by total quantity sold.

```
select od.productid, p.productname, count(*) as total_quantity_sold from
order_details od
join products p on p.productid = od.productid
group by od.productid, p.productname order by total_quantity_sold desc limit
10;
```

	123 productid	A-Z productname	123 total_quantity_sold
1	59	Raclette Courdavault	54
2	24	Guaraná Fantástica	51
3	31	Gorgonzola Telino	51
4	60	Camembert Pierrot	51
5	56	Gnocchi di nonna Alice	50
6	62	Tarte au sucre	48
7	41	Jack's New England Clam Chowder	47
8	75	Rhönbräu Klosterbier	46
9	2	Chang	44
10	16	Pavlova	43

29. Find categories whose total sales are greater than 50000.

```
select p.categoryid, c.categoryname, sum(od.unitprice * od.quantity * (1-
discount)) as total_sales from order_details od
join products p on p.productid = od.productid
join categories c on c.categoryid = p.categoryid
group by p.categoryid, c.categoryname
having sum(od.unitprice * od.quantity * (1-discount)) > 50000;
```

	123 categoryid ▼	A-Z categoryname ▼	123 total_sales ▼
1	8	Seafood	131,261.7373446997
2	2	Condiments	106,047.0848785442
3	6	Meat/Poultry	163,022.3593920469
4	5	Grains/Cereals	95,744.5874278888
5	4	Dairy Products	234,507.2847417593
6	3	Confections	167,357.2248428258
7	1	Beverages	267,868.1797437187
8	7	Produce	99,984.5799156859

30. Find employees whose total sales are greater than 100000.

```
select o.employeeid, e.firstname, sum(od.unitprice * od.quantity * (1-discount)) as
total_sales from order_details od
join orders o on o.orderid = od.orderid
join employees e on e.employeeid = o.employeeid
group by o.employeeid, e.firstname
having sum(od.unitprice * od.quantity * (1-discount)) > 100000;
```

	123 employeeid ▼	A-Z firstname ▼	123 total_sales ▼
1	4	Margaret	232,890.8457361288
2	3	Janet	202,812.8428513777
3	8	Laura	126,862.2774064841
4	7	Robert	124,568.234814424
5	1	Nancy	192,107.6043706772
6	2	Andrew	166,537.7548349826

31. Find total sales per country based on customer country.


```

select c.country as customer_country, sum(od.unitprice * od.quantity * (1-
discount)) as total_sales from order_details od
join orders o on o.orderid = od.orderid
join customers c on c.customerid = o.customerid
group by c.country;

```

	A-Z customer_country ▼	123 total_sales ▼
1	Argentina	8,119.1
2	Spain	17,983.1999947214
3	Switzerland	31,692.6589715903
4	Italy	15,770.1549791459
5	Venezuela	56,810.6289418125

32. Find suppliers whose products generated sales above 30000.

→select suppliers from products table by finding the customers in order_details table by calculating the total sales amount and filtering which customers have total sales more than 30000.

```

select p.supplierid, sum(od.unitprice * od.quantity * (1-discount)) as total_sales
from order_details od
join products p on p.productid = od.productid
join suppliers s on s.supplierid = p.supplierid
group by p.supplierid
having sum(od.unitprice * od.quantity * (1-discount)) > 30000;

```


	123 supplierid	123 total_sales
1	29	61,587.5699178308
2	4	30,526.339953503
3	7	106,459.7753825933
4	15	43,141.5099571396
5	12	145,372.3994297538
6	26	50,254.6099595234
7	24	65,626.7699364409

33. Find customers who placed more than 10 orders and sort by order count descending.

→select customers from orders table who has the total num of order more than 10, grouping by customerid from orders table and ordering in descending order by total num of sales

```
select o.customerid, c.contactname as customer_name, count(*) as total_orders
from orders o
join customers c on c.customerid = o.customerid
group by o.customerid, c.contactname
having count (*) > 10
order by count(*) desc;
```

	A-Z customerid	A-Z customer_name	123 total_orders
1	SAVEA	Jose Pavarotti	31
2	ERNSH	Roland Mendel	30
3	QUICK	Horst Kloss	28
4	FOLKO	Patricia McKenna	19
5	HUNGO	Carlos Hernández	19
6	HILAA	Paula Wilson	18
7	RATTC	Christina Berglund	18
8	BERGS	Laurence Lebihan	17
9	BONAP	Renate Messner	15
10	LEHMS		

34. Find monthly order count for each year and month.

→ finding monthly order count from orders table by grouping year and month of order date and ordering in ascending order by year and month of orderdate.

```
select extract(year from orderdate) as year, extract(month from orderdate) as month, count(*) as total_orders
from orders
group by extract(year from orderdate), extract(month from orderdate)
order by year, month;
```

	123 year ▼	123 month ▼	123 total_orders ▼
1	1,996	7	22
2	1,996	8	25
3	1,996	9	23
4	1,996	10	26
5	1,996	11	25
6	1,996	12	31
7	1,997	1	33
8	1,997	2	29
9	1,997	3	30
10	1,997	4	31

35. Find monthly sales amount ordered by year and month.

→ finding monthly sales amount from order details table by comparing ordered between orders and order details table and grouping by year and month, and ordering in ascending order by year and month of orderdate.

```
select extract(year from o.orderdate) as year,
extract(month from o.orderdate) as month,
sum(od.unitprice * od.quantity * (1-od.discount)) as total_sales
from order_details od
join orders o on o.orderid = od.orderid
group by extract(year from o.orderdate), extract(month from o.orderdate)
order by extract(year from o.orderdate), extract(month from o.orderdate);
```

	123 year ▼	123 month ▼	123 total_sales ▼
1	1,996	7	27,861.8949565172
2	1,996	8	25,485.2749865815
3	1,996	9	26,381.3999884874
4	1,996	10	37,515.7249144845
5	1,996	11	45,600.0449431259
6	1,996	12	45,239.629965432
7	1,997	1	61,258.0699304849
8	1,997	2	38,483.6349429429
9	1,997	3	38,547.2199698895
10	1,997	4	53,032.9524269359